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APPENDICES

Appendix A Calibration Curves

Table A1 Gas chromatograph's calibration curve for hydrogen (H_2)

Volume of hydrogen (ml)	Peak area
0.02	1,101,005
0.04	2,016,179
0.08	3,680,042
0.1	5,675,328
0.2	11,471,761
0.4	22,832,569

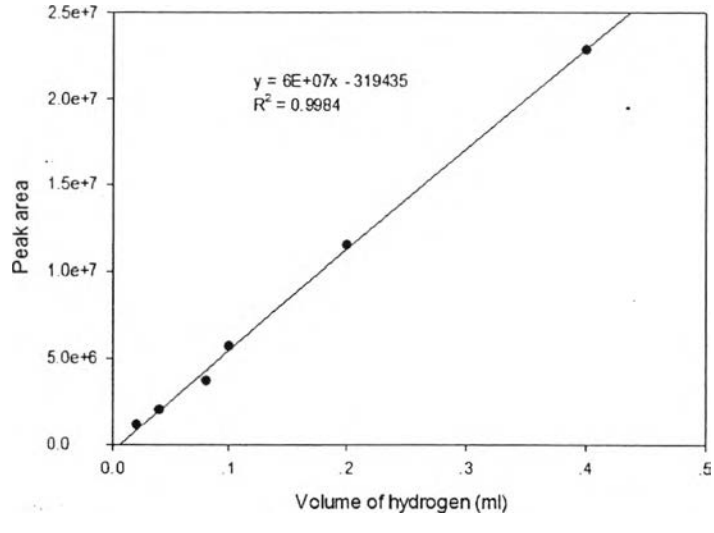


Figure A1 The relationship between amount of hydrogen (H_2) and peak area.

Equation

$$\text{Amount of hydrogen} = \frac{\text{Peak area} + 319{,}435}{6 \times 10^7}$$

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Table A2 Gas chromatograph's calibration curve for nitrogen (N_2)

Volume of nitrogen (ml)	Peak area
0.02	69,431
0.04	188,161
0.08	426,068
0.1	478,146
0.2	1,008,515
0.4	2,155,800
0.6	3,309,337

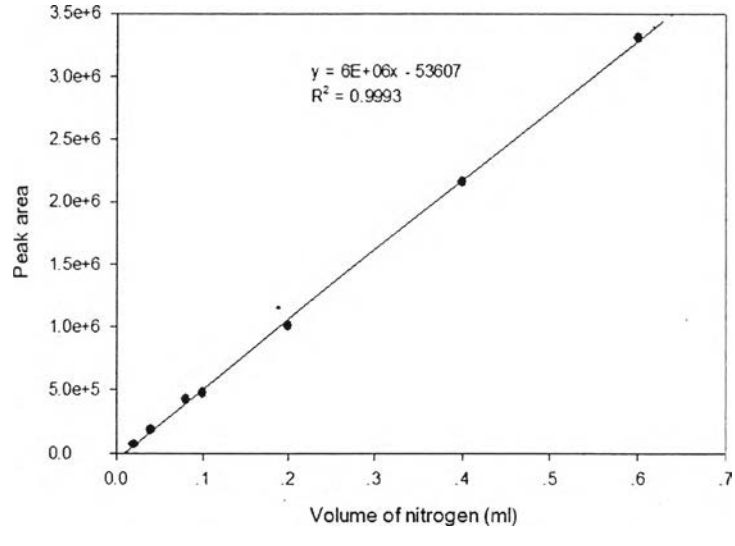


Figure A2 The relationship between amount of nitrogen (N_2) and peak area.

Equation

$$\text{Amount of nitrogen} = \frac{\text{Peak area} + 53{,}607}{6 \times 10^6}$$

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Table A3 Gas chromatograph's calibration curve for oxygen (O_2)

Volume of oxygen (ml)	Peak area
0.02	81,122
0.04	233,918
0.08	514,527
0.1	662,766
0.2	1,366,208
0.4	2,738,126

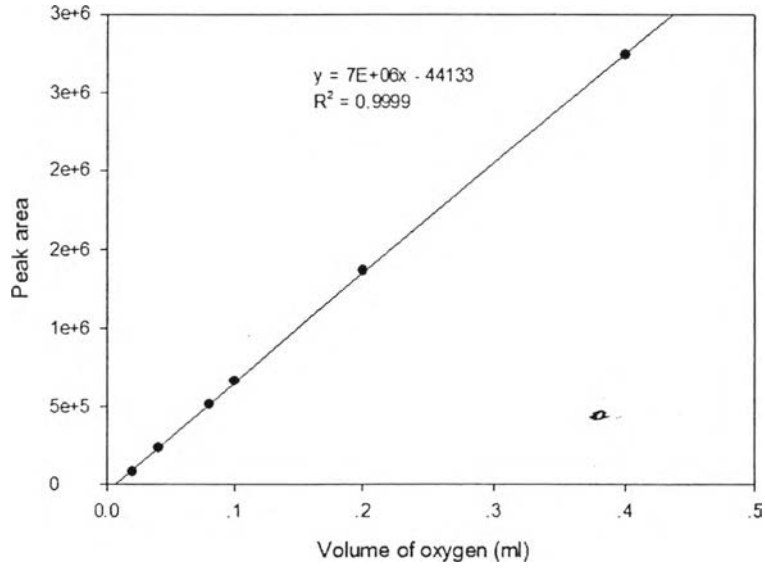


Figure A3 The relationship between amount of oxygen (O_2) and peak area.

Equation

$$\text{Amount of oxygen} = \frac{\text{Peak area} + 44{,}133}{7 \times 10^6}$$

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Table A4 Gas chromatograph's calibration curve for methane (CH_4)

Volume of methane (ml)	Peak area
0.02	151,094
0.04	523,919
0.08	998,851
0.1	1,366,651
0.2	2,898,103
0.4	5,880,444

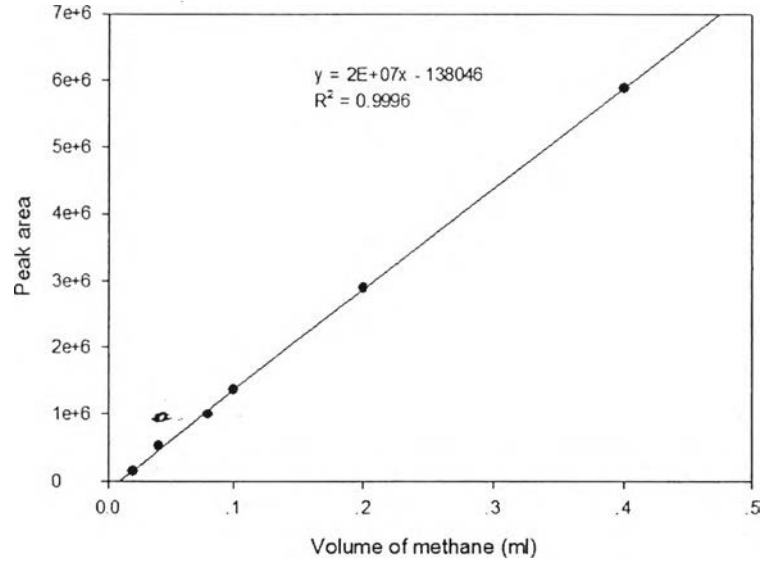


Figure A4 The relationship between amount of methane (CH_4) and peak area.

Equation

$$\text{Amount of methane} = \frac{\text{Peak area} + 138{,}046}{2 \times 10^7}$$

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Table A5 Gas chromatograph's calibration curve for carbon dioxide (CO_2)

Volume of carbon dioxide (ml)	Peak area
0.02	4,238
0.04	188,166
0.08	293,029
0.1	354,304
0.2	747,872
0.4	1,515,064

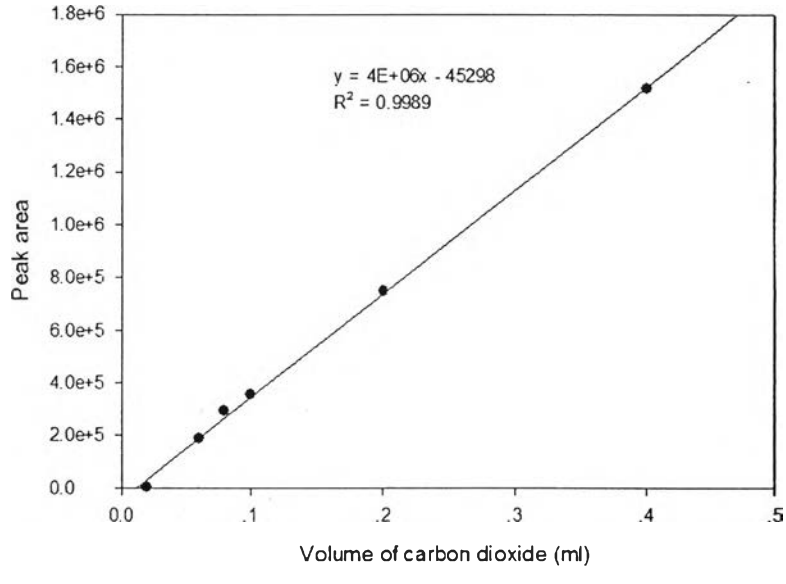


Figure A5 The relationship between amount of carbon dioxide (CO_2) and peak area.

Equation

$$\text{Amount of carbon dioxide} = \frac{\text{Peak area} + 45{,}298}{4 \times 10^6}$$

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Table A6 Liquid chromatograph's calibration curve for acetic acid

Concentration of acetic acid (mg/l)	Peak area
1,000	226,593
2,000	458,639
3,000	693,445
4,000	951,778
5,000	1,179,161

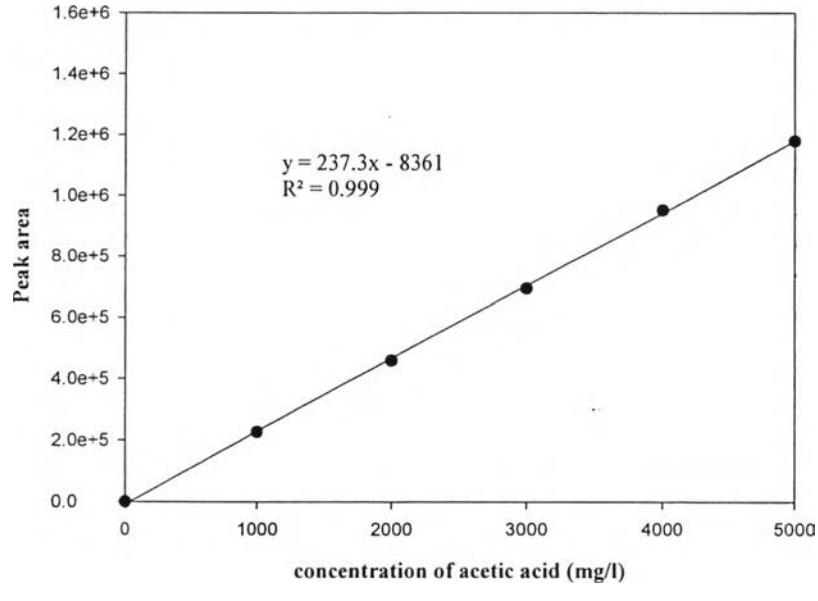


Figure A6 The relationship between concentration of acetic acid and peak area.

Equation

$$Amount\ of\ acetic\ acid = \frac{Peak\ area + 8361}{237.3}$$

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Table A7 Liquid chromatograph's calibration curve for propionic acid

Concentration of propionic acid (mg/l)	Peak area
1,000	274,670
2,000	553,990
3,000	836,683
4,000	1,091,859
5,000	1,435,669

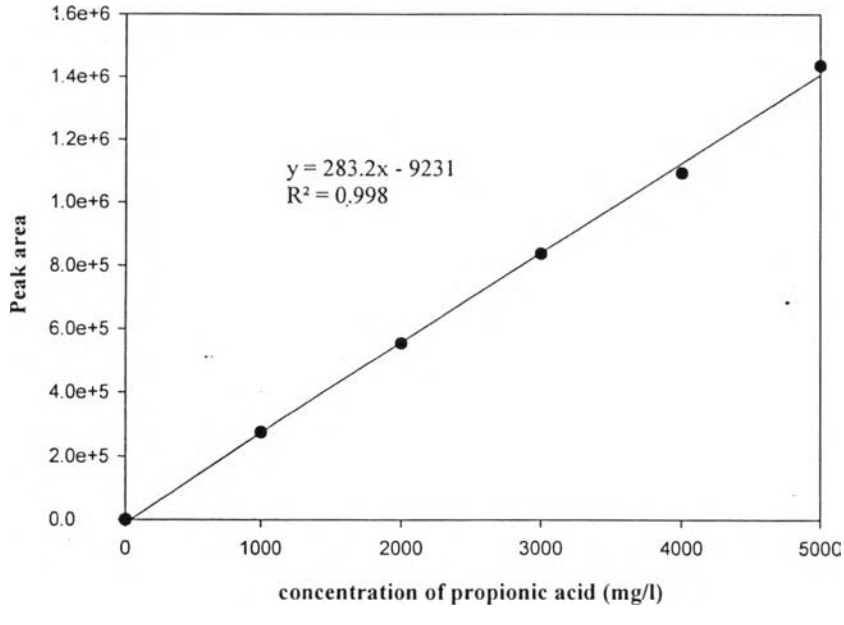


Figure A7 The relationship between concentration of propionic acid and peak area.

Equation

$$Amount\ of\ propionic\ acid = \frac{Peak\ area + 9231}{283.2}$$

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Table A8 Liquid chromatograph's calibration curve for butyric acid

Concentration of butyric acid (mg/l)	Peak area
1,000	310,185
2,000	636,623
3,000	974,830
4,000	1,315,752
5,000	1,672,791

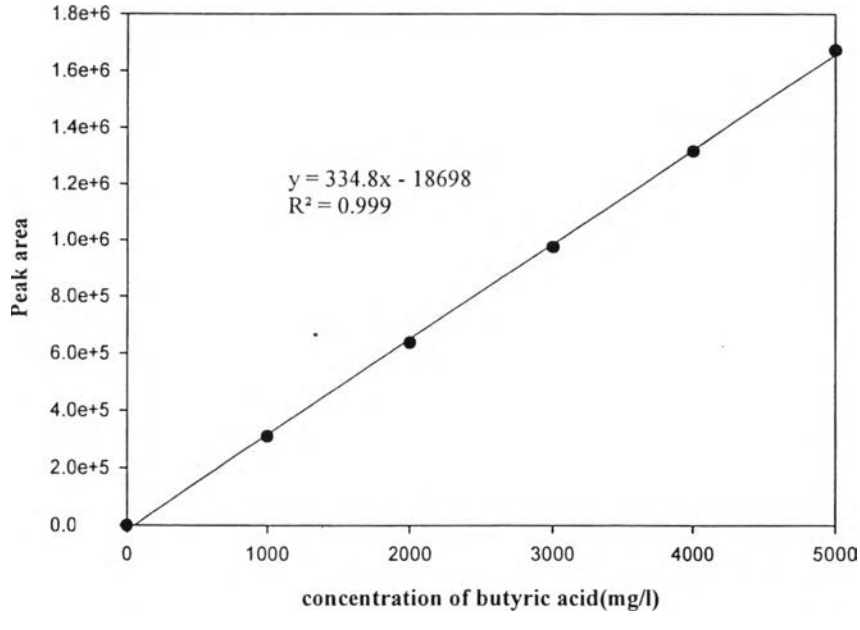


Figure A8 The relationship between concentration of butyric acid and peak area.

Equation

$$Amount\ of\ butyric\ acid = \frac{Peak\ area + 18698}{334.8}$$

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Table A9 Liquid chromatograph's calibration curve for valeric acid

Concentration of valeric acid (mg/l)	Peak area
1,000	346,808
2,000	707,645
3,000	1,082,011
4,000	1,470,955
5,000	1,844,040

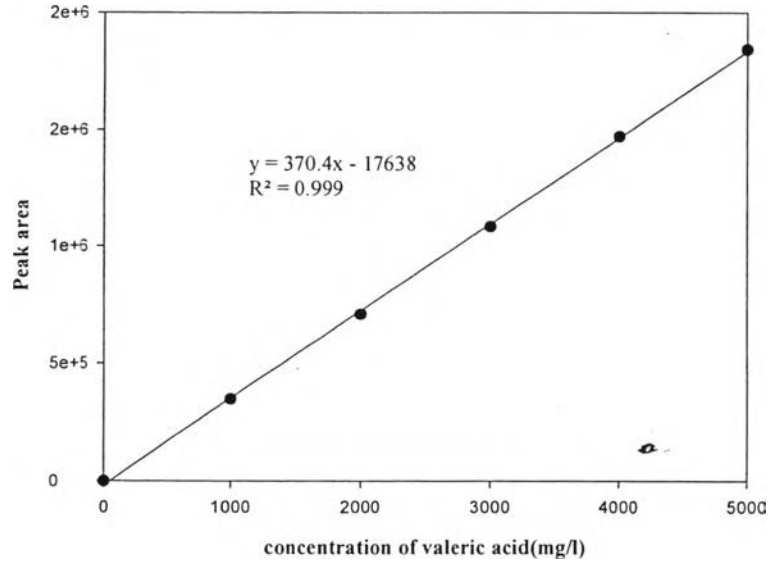


Figure A9 The relationship between concentration of valeric acid and peak area.

Equation

$$Amount\ of\ valeric\ acid = \frac{Peak\ area + 17638}{370.4}$$

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Table A10 Liquid chromatograph's calibration curve for ethanol

Concentration of ethanol (mg/l)	Peak area
1,000	189,866
2,000	377,275
3,000	569,223
4,000	765,786
5,000	958,108

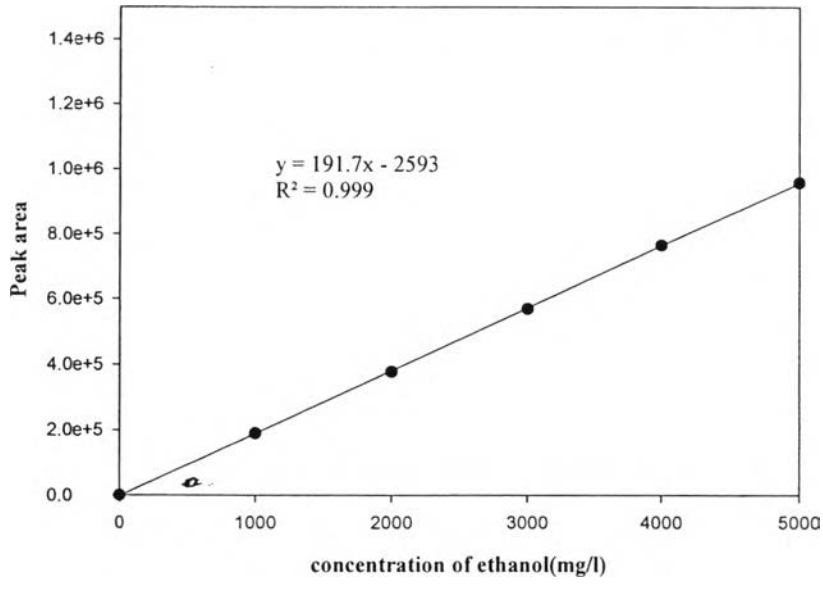


Figure A10 The relationship between concentration of ethanol and peak area.

Equation

$$Amount\ of\ et\square anol = \frac{Peak\ area + 2593}{191.7}$$

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Table A11 Liquid chromatograph's calibration curve for lactic acid

Concentration of ethanol (mg/l)	Peak area
1,000	241,057
2,000	513,754
3,000	735,330
4,000	1,047,749
5,000	1,350,707

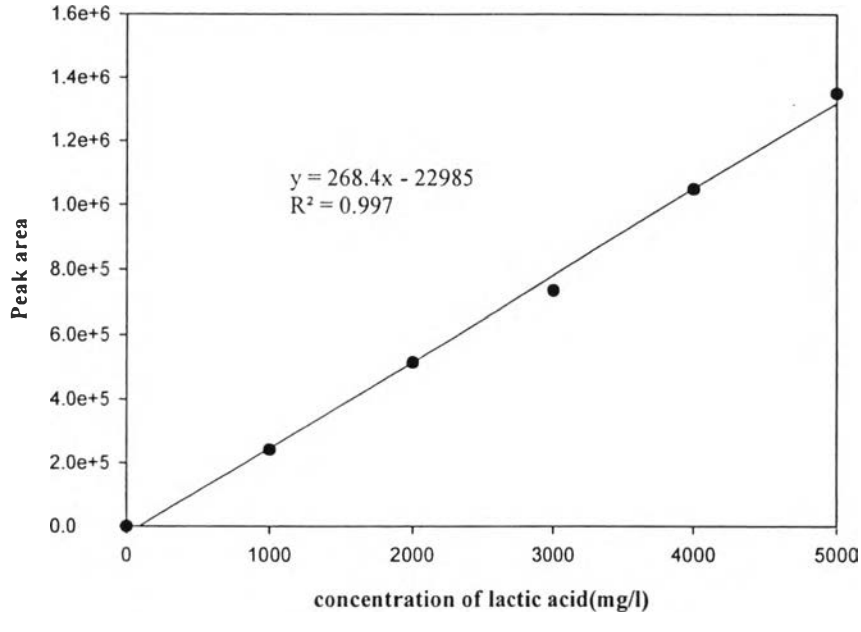


Figure A11 The relationship between concentration oflacticand peak area.

Equation

$$Amount\ of\ et\square anol = \frac{Peak\ area + 22985}{268.4}$$

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Appendix B Preparation of 1 MNaOH Solution for pH-controlled System

Preparation of NaOH at concentration of 1 M

(Molecular weight of acetic acid = 60)

$$= \frac{1\ mol}{1\ l} \times \frac{40\ g}{1\ mol}$$

$$= 40 \frac{g}{l}$$

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Appendix C Volatile Fatty Acids (VFA) Quantification by using High-Performance Liquid Chromatography

C 1. Mobile phase solution preparation

Concentration of fresh sulfuric acid (liquid) = 98%v/v

Density of sulfuric acid = 1.84 g/ml

Molecular weight of sulfuric acid = 98.08

Determination of fresh acetic acids concentration in term of molar

$$= \frac{0.98 mL \text{ of sulfuric acid}}{\text{mL of solution}} \times \frac{1.84 \text{ g of sulfuric acid}}{\text{mL of sulfuric acid}} \times \frac{1 \text{ mol of sulfuric acid}}{98.08 \text{ g of sulfuric acid}} \times \frac{1000 mL}{1 L}$$

$$= 18.38 \text{ M}$$

Sulfuric acid are required at concentration of 0.004M

Dilution of sulfuric acid

$N_1V_1 = N_2V_2$

$V_1 = N_2V_2/N_1$

$= (0.004x2)/18.38$

$= 4.352 \times 10^{-4}$ L

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Proceedings:

1. Tuntiwichayanon, P.; and Chavadej, S. (2015, April 21) Effects of Microaeration on Hydrogen and Methane Production from Cassava Wastewater using a Two-stage UASB System. Proceedings of The6thResearch Symposium on Petroleum, Petrochemicals, and Advanced Materials and The 21th PPC Symposium on Petroleum, Petrochemicals, and Polymers, Bangkok, Thailand.

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